# Chapter 5. Potential Problems in Regression

Lecture 6-2 (assumptions, non-Gaussianity, outliers and robust regression, breakdown point, measurement error)

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# 1 Assumptions and Common Issues

The linear model relies on: linearity; normally distributed errors; independent errors; constant variance; no outliers or high-leverage points; small collinearity.

**Outliers** are observations with unusually large residuals; **high-leverage points** have unusual predictor values; **correlated errors** violate independence.

| Kind of problem | Examples |
|---|---|
| data problems | outliers, measurement error, missing data |
| model problems | omitted-variable bias, misspecification, multicollinearity |
| error problems | non-normality, heteroscedasticity, autocorrelation |

**Remark: diagnostics in brief**

**Nonlinearity**: residuals versus fitted values show a pattern (fix: transformations, polynomials, splines). **Heteroscedasticity**: a funnel in the residual plot (fix: transform $Y$, e.g. $\log Y$ or $\sqrt{Y}$, or weighted least squares). **Correlated errors**: tracking in residuals over time; standard errors become too small. **Leverage**: $h_{ii} = x_i^\top (X^\top X)^{-1} x_i$, average $(p+1)/n$; flag $h_{ii} \gg (p+1)/n$. **Collinearity**: variance inflation factor $\mathrm{VIF}(\hat{\beta}_j) = 1/(1 - R^2_{X_j | X_{-j}})$; values above 5–10 are problematic.

# 2 Non-Gaussianity: Normality Is Not an Issue

**Theorem: exact distribution under Gaussian errors**

If $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I)$ and $X$ of full column rank, then

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^\top X)^{-1}), \qquad \hat{\beta}_j \sim N(\beta_j, \sigma^2 v_j),$$

where $v_j$ is the $j$th diagonal element of $(X^\top X)^{-1}$. Replacing $\sigma$ by $\hat{\sigma} = \sqrt{\mathrm{RSS}/(n-p)}$ ($p$ columns of $X$ including the intercept),

$$t_j = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{v_j}} \sim t_{n-p},$$

which gives tests and confidence intervals.

**Proof.** $\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon$ is a linear map of a Gaussian vector, with mean $\beta$ and covariance $\sigma^2 (X^\top X)^{-1}$. The residual vector $(I - H)\varepsilon$ is independent of $\hat{\beta}$ (uncorrelated jointly Gaussian), and $\mathrm{RSS}/\sigma^2 = \varepsilon^\top (I-H)\varepsilon/\sigma^2 \sim \chi^2_{n-p}$ since $I - H$ is a projection of rank $n - p$. The ratio of a standard normal and an independent $\sqrt{\chi^2_{n-p}/(n-p)}$ is $t_{n-p}$. ∎

## 2.1 Large-sample inference

If the errors are not normal, $\hat{\beta}$ is not exactly normal, but by the CLT

$$\hat{\beta} \;\dot{\sim}\; N(\beta, \sigma^2 (X^\top X)^{-1}) \quad \text{for large } n,$$

(provided errors are i.i.d. with finite variance and the design is not dominated by a few points, i.e. $\max_i h_{ii} \to 0$). Hence $t$-tests and confidence intervals remain approximately valid. Alternatively use

**bootstrap inference**: resample the data with replacement, recompute $\hat{\beta}$ for each bootstrap sample, and use the empirical distribution of $\hat{\beta}^*$ for standard errors and intervals (Ch. 3). Resampling pairs $(x_i, y_i)$ is also valid under heteroscedasticity.

# 3 Outliers and Robust Regression

## 3.1 Why OLS is sensitive

OLS minimises $\sum_{i=1}^{n}(y_i - x_i^\top \beta)^2$. Squaring residuals amplifies large errors, so a single extreme observation can strongly influence the fitted model. This motivates robust regression.

## 3.2 Alternative objectives

- **Least absolute deviation (LAD)**: $\hat{\beta} = \arg\min_\beta \sum_i |y_i - x_i^\top \beta|$; the conditional median (Ch. 2 analogue: absolute loss is minimised by the median).
- **Huber regression**: $\hat{\beta} = \arg\min_\beta \sum_i \rho_c(r_i)$, $r_i = y_i - x_i^\top \beta$,

$$\rho_c(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le c, \\ c|r| - \frac{1}{2}c^2, & |r| > c. \end{cases}$$

- **Least trimmed squares (LTS)**: $\hat{\beta} = \arg\min_\beta \sum_{i=1}^{h} r^2_{(i)}$, using only the $h$ smallest squared residuals ($r^2_{(1)} \le r^2_{(2)} \le \cdots$).

**Derivation: Huber as iteratively reweighted least squares**

$\rho_c$ is convex with derivative $\psi_c(r) = r$ for $|r| \le c$ and $c\,\mathrm{sign}(r)$ otherwise. The estimating equation $\sum_i \psi_c(r_i)x_i = 0$ equals the weighted normal equation $\sum_i w_i r_i x_i = 0$ with $w_i = \min(1, c/|r_i|)$, so one iterates $\beta \leftarrow (X^\top W X)^{-1} X^\top W y$. Large residuals get weights $c/|r_i| < 1$: their influence is bounded by $c$, whereas under OLS it is unbounded. Usually $c = 1.345\hat{s}$ for a robust scale $\hat{s}$ (95% efficiency under normal errors).

## 3.3 Breakdown point

**Definition: breakdown point**

The (finite-sample) breakdown point of an estimator is the smallest fraction of contaminated observations that can make the estimate arbitrarily large (break it down).

| OLS | LAD | Huber M-estimator | least median of squares | least trimmed squares |
|---|---|---|---|---|
| 0 | 0 | 0 | 0.5 | $\approx 0.5$ |

OLS can be destroyed by a single outlier (breakdown $1/n \to 0$): moving one $y_i$ to infinity moves $\hat{\beta}$ to infinity. LAD and Huber resist outliers in $y$, but a single bad *leverage point* (outlying $x_i$) can still break them, hence breakdown 0. High-breakdown estimators (LMS: minimise the median squared residual; LTS with $h \approx n/2$) tolerate nearly 50% contamination — more than half bad points can never be distinguished from good ones, so 0.5 is the maximum.

# 4 Measurement Error

Suppose the true model is $Y = \beta_0 + \beta_1 X + \varepsilon$, but the predictor is measured with error:

$$\tilde{X} = X + u, \qquad u \text{ independent of } (X, \varepsilon),\ \mathbb{E}u = 0.$$

Regressing $Y$ on the observed $\tilde{X}$ gives a biased estimator.

**Theorem: attenuation bias**

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 \frac{\mathrm{Var}(X)}{\mathrm{Var}(X) + \mathrm{Var}(u)} = \lambda \beta_1, \qquad 0 < \lambda \leq 1.$$

**Proof.** The OLS slope converges to $\mathrm{Cov}(\tilde{X}, Y)/\mathrm{Var}(\tilde{X})$. Here $\mathrm{Cov}(X + u, \beta_0 + \beta_1 X + \varepsilon) = \beta_1 \mathrm{Var} X$ and $\mathrm{Var}(\tilde{X}) = \mathrm{Var} X + \mathrm{Var} u$. ■

The slope is shrunk toward zero (**attenuation bias**); $\lambda$ is the reliability ratio. Measurement error in $Y$ only, in contrast, inflates the error variance but does not bias the slope. In multiple regression, error in one predictor also biases the coefficients of correlated predictors.

## 4.1 Remedies

| Method | Idea |
|---|---|
| instrumental variables | use a variable $Z$ correlated with $X$ but independent of the measurement error: $\hat{\beta}_1^{\mathrm{IV}} = \widehat{\mathrm{Cov}}(Z, Y)/\widehat{\mathrm{Cov}}(Z, \tilde{X})$ |
| repeated measurements | average $m$ noisy measurements, reducing $\mathrm{Var} u$ to $\mathrm{Var} u/m$; replicates also estimate $\mathrm{Var} u$ so that $\hat{\beta}_1/\hat{\lambda}$ corrects the bias |
| errors-in-variables models | model the measurement process explicitly (e.g. Deming regression) |
| SIMEX | add extra artificial noise of increasing size, observe how the estimate changes, and extrapolate back to zero measurement error |
| validation data | use a subset where the true $X$ is observed to calibrate $\tilde{X}$ (regression calibration) |

# Summary

- Regression problems arise from the data (outliers, measurement error, missing data), the model (omitted variables, misspecification, collinearity) and the error structure (heteroscedasticity, autocorrelation).
- Normality is often not critical: the CLT gives large-sample inference and the bootstrap is an alternative.
- Outliers motivate robust regression (LAD, Huber, LTS); the breakdown point measures robustness, and only high-breakdown methods resist leverage points.
- Error in a predictor causes attenuation bias toward zero; IV, replicates, SIMEX or validation data correct it.